

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using Py Spark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Reverse Engineering

Code of Conduct:

*I _____ (V. Asha Sarthiwika / 192311066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:V. Asha Sarthiwika

Annexure A

Reverse Engineering

1. Reverse Engineering a Recommendation System

A large-scale e-commerce platform provides highly accurate product recommendations using real-time data. Analyze the system and reverse engineer its architecture using PySpark and RDD operations. Identify how data is collected, processed, and transformed, and infer the role of intelligent agents in updating recommendations dynamically.

2. Reverse Engineering a Smart Traffic Management System

A city-wide traffic system adapts signals in real time based on vehicle density and congestion patterns. Reverse engineer the underlying AI system by describing the data pipeline using RDD-based streaming, identifying agent interactions, and explaining how decisions are made under time constraints.

3. Reverse Engineering a Fraud Detection Pipeline

A banking system flags fraudulent transactions within milliseconds. By observing its outputs and behavior, reconstruct the distributed AI pipeline using PySpark RDDs. Explain how feature extraction, anomaly detection, and agent-based alert mechanisms are likely implemented.

4. Reverse Engineering a Personalized Learning System

An online learning platform adapts content difficulty based on student performance and engagement. Reverse engineer the system design by identifying how learner data is processed using RDD transformations, and explain how intelligent agents contribute to adaptive learning and feedback mechanisms.

5. Reverse Engineering a Cloud-Based AI Monitoring System

A cloud platform automatically scales resources and detects system anomalies in real time. Reverse engineer the architecture by analyzing how PySpark processes large-scale logs, how RDD operations support fault tolerance, and how intelligent agents coordinate monitoring, scaling, and anomaly response.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. Reverse Engineering a Recommendation System

A recommendation system collects user data such as searches, product views, ratings, and purchases. PySpark RDDs are used to process this large amount of data. Operations like `map()`, `filter()`, `groupByKey()`, and `reduceByKey()` help organize and analyze user and product information. The processed data is used to identify customer preferences and recommend suitable products. An intelligent agent continuously observes user behavior and updates recommendations when the user's interests change.

Flow:

User Data → RDD Processing → Preference Analysis → Recommendation → Agent Update

2. Reverse Engineering a Smart Traffic Management System

A smart traffic system collects real-time information from cameras, sensors, and traffic signals. PySpark RDDs process vehicle density and congestion data from different roads. RDD operations help filter, group, and analyze traffic information quickly. Intelligent agents monitor traffic conditions and communicate with traffic signals. Based on the current traffic level, the agent can increase or decrease signal timing to reduce congestion.

Flow:

Traffic Sensors → RDD Processing → Traffic Analysis → Intelligent Agent → Signal Control

3. Reverse Engineering a Fraud Detection Pipeline

A fraud detection system collects transaction details such as amount, location, time, and transaction type. PySpark RDDs process these transactions and extract important features. The system compares normal and unusual transaction patterns to detect possible fraud. When suspicious activity is found, an intelligent agent

generates an alert and can block or verify the transaction. This allows banks to detect fraud quickly.

Flow:

Transaction Data → RDD Processing → Feature Extraction → Fraud Detection → Alert Agent

4. Reverse Engineering a Personalized Learning System

A personalized learning system collects student information such as marks, quiz results, completed lessons, and learning time. PySpark RDDs process this data to understand student performance and learning behavior. The system identifies whether a student is performing well or needs additional support. An intelligent agent then selects suitable learning materials and changes the difficulty level. It also provides feedback based on the student's progress.

Flow:

Student Data → RDD Processing → Performance Analysis → Intelligent Agent → Personalized Learning

5. Reverse Engineering a Cloud-Based AI Monitoring System

A cloud monitoring system collects server logs, CPU usage, memory usage, network activity, and error information. PySpark RDDs process these large-scale logs and identify unusual system behavior. RDDs provide distributed and fault-tolerant processing, so data can continue to be processed even if a worker fails. Intelligent agents monitor the system, detect anomalies, automatically increase or decrease resources, and respond to failures.

Flow:

Cloud Logs → RDD Processing → Anomaly Detection → Intelligent Agent → Scaling and Response